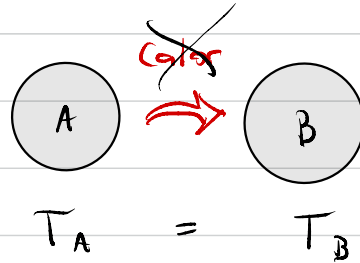
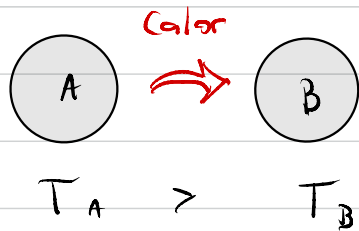


Leis da Termodinâmica

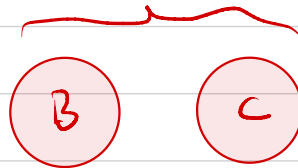
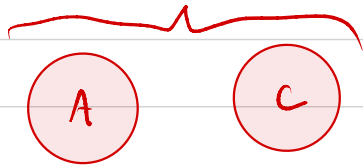
• Lei Zero $\approx$ Equilíbrio Térmico



• O calor está ~~contido~~ num calor

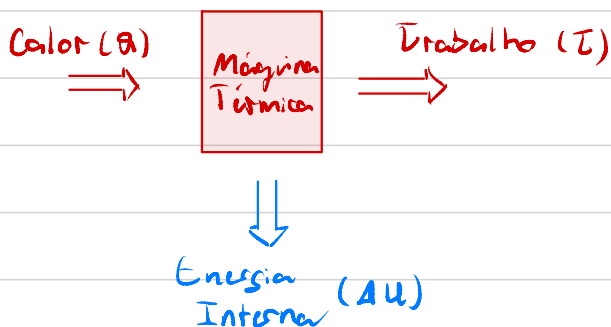
• O calor sempre flui de $\uparrow T$ para $\downarrow T$ espontaneamente.

(A) está em equilíbrio térmico com (B)



Se $T_A = T_C$ e também $T_B = T_C$

• Primeira Lei da Termodinâmica $\approx$ Conservação de Energia

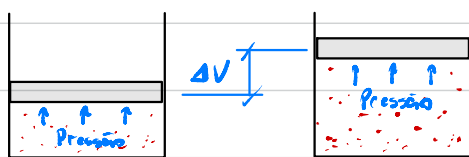
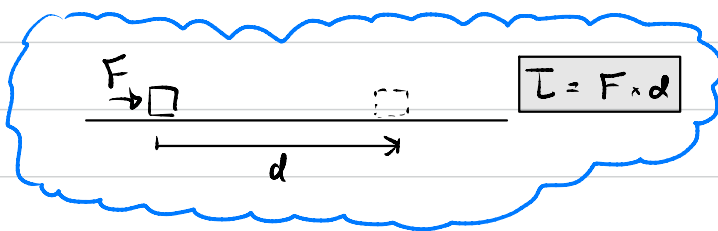


• $Q = L + \Delta U$ ✓

ou

• $\Delta U = Q - L$

Trabalho de um gás $\approx$ Volume!



$$W_{\text{gás}} = P \times \Delta V_{\text{li}}$$

• Transf. Isovolumétrica

$$\Delta V_{\text{li}} = 0 \Rightarrow W = P \times \cancel{\Delta V} = 0$$

• Expansão

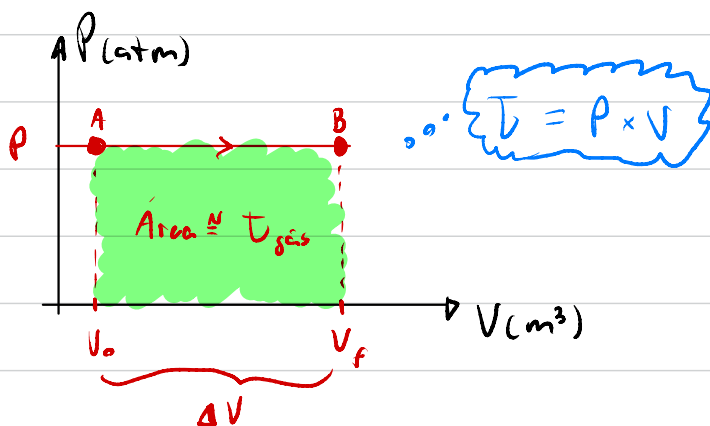
$$V_f > V_o \rightarrow \Delta V \oplus \rightarrow W \oplus$$

• Contração

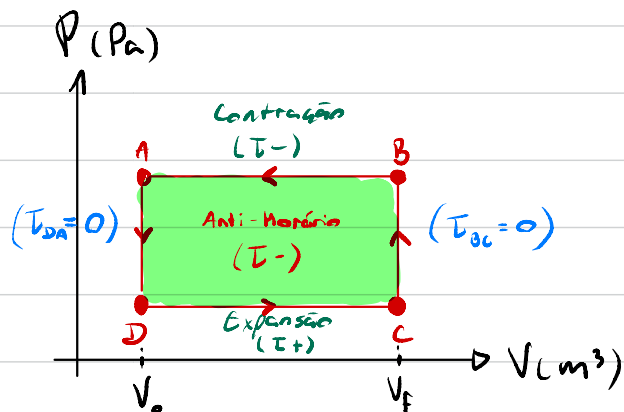
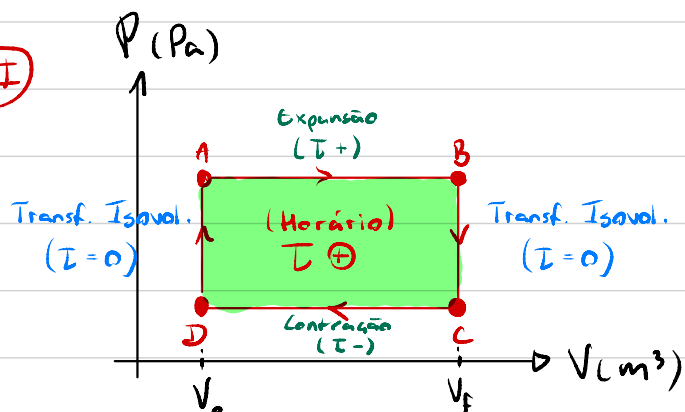
$$V_f < V_o \rightarrow \Delta V \ominus \rightarrow W \ominus$$

• $W_{\text{gás}}$ na gráfica $P \times V$

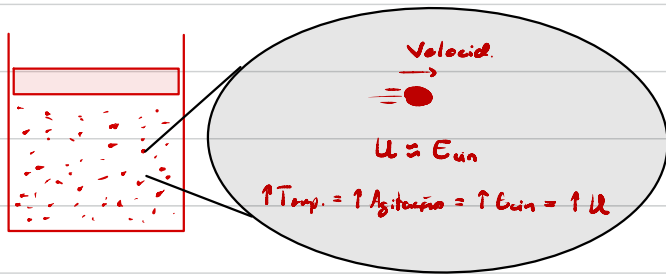
(I)



(II)



Energia Interna $\approx$ Temperatura!



i) Gases Monatômicos

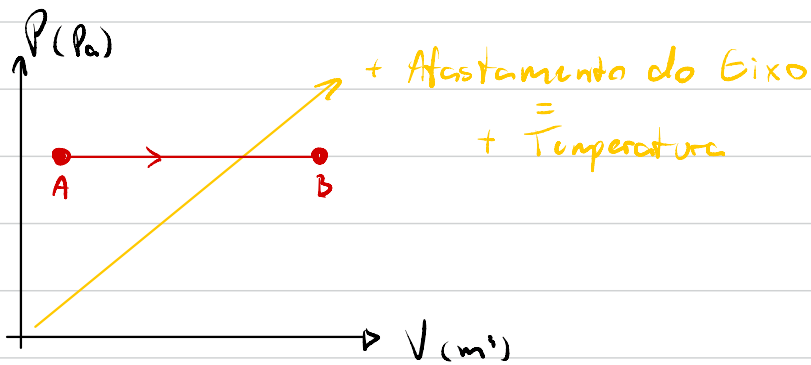
(90%)

$$U = \frac{3}{2} nRT$$

ii) Gases Diatômicos

$$U = \frac{5}{2} nRT$$

• Energ. Interna no Gráfico $P \times V$



$$\left. \begin{array}{l} U_B > U_A \\ T_B > T_A \end{array} \right\} \Delta U \oplus$$

Calor (Q)

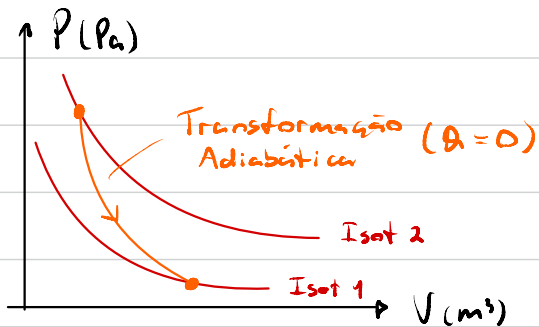
$\text{Calor } Q = m \cdot c \cdot \Delta T$

$\rightarrow Q_p = m \cdot c_p \cdot \Delta T$ [Isobárica]

$\rightarrow Q_v = m \cdot c_v \cdot \Delta T$ [Isotérmica]

* Calor no Gráfico P x V

i) Transformação Adiabática



ii) Outras Transformações

